Megathon X — Qualcomm Track | Milestone 1: Technical Approach

Team: B612 | **Project:** End-to-End On-Device AI Incident Report Assistant | **Hardware:** Snapdragon® 8 Gen 3 (HDK8650)

1. Problem Statement & Deep Architectural Understanding

Traditional enterprise incident logging (safety hazards, infrastructure failures, chemical spills) relies on tedious multi-field web/mobile forms that workers delay or fail to complete under pressure. Qualcomm's Megathon X challenge mandates replacing manual entry with a **voice-driven, hands-free conversational assistant** that extracts structured information from natural narration.

Core Constraints & Official Requirements:

- **Zero Cloud Execution (Mandatory):** All models (Speech-to-Text, LLM extraction, and TTS synthesis) must execute strictly on-device utilizing Snapdragon NPU / DSP acceleration via Qualcomm AI Runtime (QAIRT / QNN). Cloud APIs are disqualified.
- **Strict 10-Field Data Contract:** Schema requires extracting 4 Mandatory fields (incident_type, incident_location, date_time, immediate_actions_taken) and 6 Optional fields (severity_level, persons_involved, equipment_asset_involved, incident_description, witness_information, recommended_follow_up_action).
- **Interactive Diamond Gate:** Missing mandatory fields must trigger an active conversational clarification loop asking single targeted questions rather than dumping errors. Once all 4 mandatory fields are satisfied, the system emits a tool call (populate_incident_form) for UI hydration and review.

2. Idea & Abstract

We present **IncidentIQ On-Device**: a resilient edge-intelligence architecture running on the Qualcomm HDK8650 (SM8650P). The system couples **Snapdragon DSP-accelerated Whisper ASR** with a quantized **Llama-3.2 / SLM information extractor running on the Hexagon NPU via Genie/QAIRT**. Rather than relying on simple keyword matching or fragile one-shot prompting, our core innovation is a **Deterministic State-Machine Agent Brain** that preserves conversational history across turns, defends against SLM hallucinations via multi-layer JSON sanitation, and guarantees zero cloud dependency with instantaneous sub-second audio turnarounds.

Field Name	Status	Extraction Logic / Description
incident_type	**MANDATORY**	Extracted or classified (e.g., Water Leak, Chemical Spill, Power Outage, Injury)
incident_location	**MANDATORY**	Physical facility, floor, lab, room, or perimeter identified from narration
date_time	**MANDATORY**	Timestamp or temporal reference normalized into structured time format
immediate_actions_taken	**MANDATORY**	Containment steps executed (e.g., valve shutoff, evacuation, area cordoned)
6 Optional Fields	Optional	Severity (Low-Critical), personnel, equipment, description, witnesses, follow-up

3. High-Level Pipeline & System Architecture

Our solution operates as a synchronized hardware-software pipeline bridging physical Snapdragon NPU execution with an intuitive live incident operations dashboard:

Stage	Engine & Hardware Layer	Functionality & Data Flow
1. Audio Capture	Host / On-Device Mic (16kHz PCM WAV)	Captures voice notes with VAD (Voice Activity Detection) / Push-to-Talk.
2. Speech-to-Text	Whisper on Snapdragon DSP / NPU	Whisper DLC encoder running on DSP + TFLite decoder transcribes speech locally.
3. State Engine Brain	State Engine (agent_brain.py)	Constructs structured prompts incorporating conversation history and known fields.
4. SLM Extraction	Llama-3.2 on Hexagon NPU (Genie/QAIRT)	NPU generates strict JSON schema output; parsed through regex-defensive parsers.
5. Diamond Gate	Deterministic Decision Logic	Evaluates the 4 mandatory fields. If missing -> status: collecting with 1-sentence clarification. If present -> status: complete.
6. Tool Call & UI	FastAPI Server & Streamlit Dashboard	Emits populate_incident_form, lights form fields green, renders transcript.
7. Offline TTS	Offline TTS Engine (Device Speaker)	Speaks clarification prompts or final incident summary aloud.

4. Planned Testing & Validation Methodology

Our quality assurance framework guarantees robust performance under real-world industrial environments across 4 rigorous testing pillars:

- **Automated Integration & Contract Tests:** Complete Pytest test suite (currently 36 passing tests) validating every state transition, JSON sanitization edge case, illegal severity handling, and multi-turn conversational memory persistence.
- **Diamond Gate Multi-Turn Dialogue Scenarios:** Verified two-phase recovery loops (e.g., Turn 1: *'A pipe burst on the second floor'* -> Prompt: *'At what time did this occur?'* -> Turn 2: *'At 2 PM and we isolated the valve'* -> Immediate form completion without overwriting existing data).
- **NPU Hardware Bridge & Fallback Resiliency:** Validated over ADB TCP forwarding (adb forward tcp:8080 tcp:8080). Includes an automated device-health watchdog with 30s cooldown and deterministic local fallback so live judging demonstrations never stall.
- **Acoustic & Noise Stress Testing:** Evaluates Whisper STT accuracy under noisy ambient industrial background sounds (machinery hum, room reverberation, informal speech pacing).

Deliverable Milestone: M1 (Approach Document) | **Repository:** IIIT-ECell/megathon26-b612 | **Date:** October 10, 2026